Design-to-Manufacture Workflows of Sound-Scattering Acoustic Brick Walls

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1

ABSTRACT

Improving speech intelligibility in classrooms enhances information dissemination, institutional knowledge capture, and quality of learning experience. While off-the-shelf solutions are available for acoustically retrofitting existing learning spaces, they do not allow for a fine-tuned context-specific intervention. However, this possibility is enabled through bespoke digital manufacturing informed by advanced digital simulations. In this research we explore and synchronize architecture, acoustics, computation, and fabrication for the making of better sound environments.

We present performance-driven design-to-manufacture (DTM) workflows for sound-scattering brick elements. We reimagine the brick as an acoustically active geometry capable of modulating the sound experience in a university classroom by improving speech intelligibility. We contextualize our research within existing methods of digital performance-based design and robotic fabrication processes, namely wire cutting and pick-and-place applications. We then detail digital methods that combine heuristics and acoustic simulation to design the bricks within the 3D modeling environment, as well as describe the processes of robotic oscillating wire cutting and adaptive pick-and-place developed for the execution of the full-scale demonstrator. Finally, we report on the results of the acoustic analysis performed on the full-scale demonstrator in situ and laboratory measurements of a representative demonstrator which validates our design hypothesis.

1 Close-up of the full-scale demonstrator with the Cusp brick family variants in focus.

INTRODUCTION

The use of earthen bricks in architecture is a millenary craft. In the preindustrial era, the adoption of mud bricks to build simple stacked architectural enclosures was limited to hot climate areas, since they were hand-molded and air-dried. The discovery of firebricks in Roman times led to the proliferation of the fired brick as a building material. Firing allowed the material to acquire a better compressive strength from its unfired counterparts, which led to the development of new structural forms such as arches and domes previously impossible with mud bricks. With the advent of industrialization, brick manufacturing lent itself well to large-scale production chains. Wet clay could be continuously extruded through a die, and the development of new kilns such as the Hoffman kiln or the Tunnel kiln guaranteed a continuous firing production line. These advancements made bricks an affordable construction material that leveraged handcraft knowledge to produce beautiful buildings (Campbell and Pryce 2016). Historically, the main driver for the shape of the brick, beyond its structural function, was the size of the human hand and the weight it could comfortably carry, considering all bricklaying was performed manually. Bricks are sized so that their lateral dimensions, with the inclusion of a mortar joint, are multiples of each other. This allows for the bricks to be arranged in interlocking patterns, known as bonds, which have both a structural performance as well as an aesthetic expression (Hall and Cruickshank 2019). Yet, since the postwar era and with the rise of the cost of handcraft, we see less and less on-site building of masonry walls. In fact, most contemporary facades that include bricks are prefabricated (Krechting 2004). These systems favor efficiency and cost reduction to the detriment of tectonics and aesthetics.

This project looks at the humble brick from a distinct perspective. Rather than thinking of the brick as a structural module, we assign to it a different performance: that of acoustic modulation. By doing so, we examine the intimate relationship between design, performance, and manufacturing in the age of digital simulation and robotics. In fact, bricks could be well-suited to bespoke acoustic treatments: they are cheap to produce and relatively lightweight, and their porous texture can provide sound modulating qualities on a micro scale. However, to be diffusive at a macro scale, bricks must move away from a prismatic profile to new custom shapes that better respond to the acoustic setting of their environment. This cannot be achieved via die extrusion, considering changes in design can become very costly and nonscalable. However, it could be achieved by leveraging digital 4.0 technology with existing industrial-scale manufacturing.

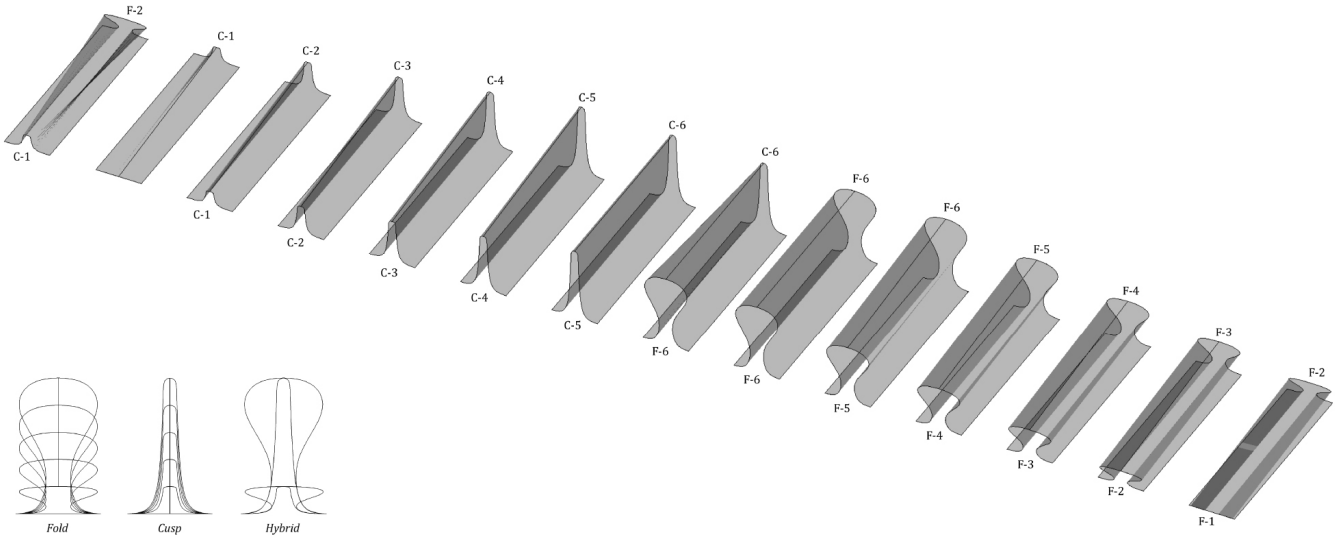
We propose novel design-to-manufacture workflows that embed acoustic performance, a clay material system, and robotic fabrication limits within the design of the brick. We demonstrate the deployment of this workflow with our industrial partner, Strøjer Tegl, in the making of a full-scale acoustic wall demonstrator. We first review examples of state-of-the-art performance-driven brick design and manufacture. Then we unpack our methodology for the design of the bricks using acoustic simulation software and their manufacture combining novel robotic oscillating wire cutting and adaptive robotic pick-and-place methods. Finally, we evaluate the acoustic performance of our wall and its tectonic qualities, discussing the advantages and limits of our methods.

PERFORMANCE-DRIVEN BRICK DESIGN

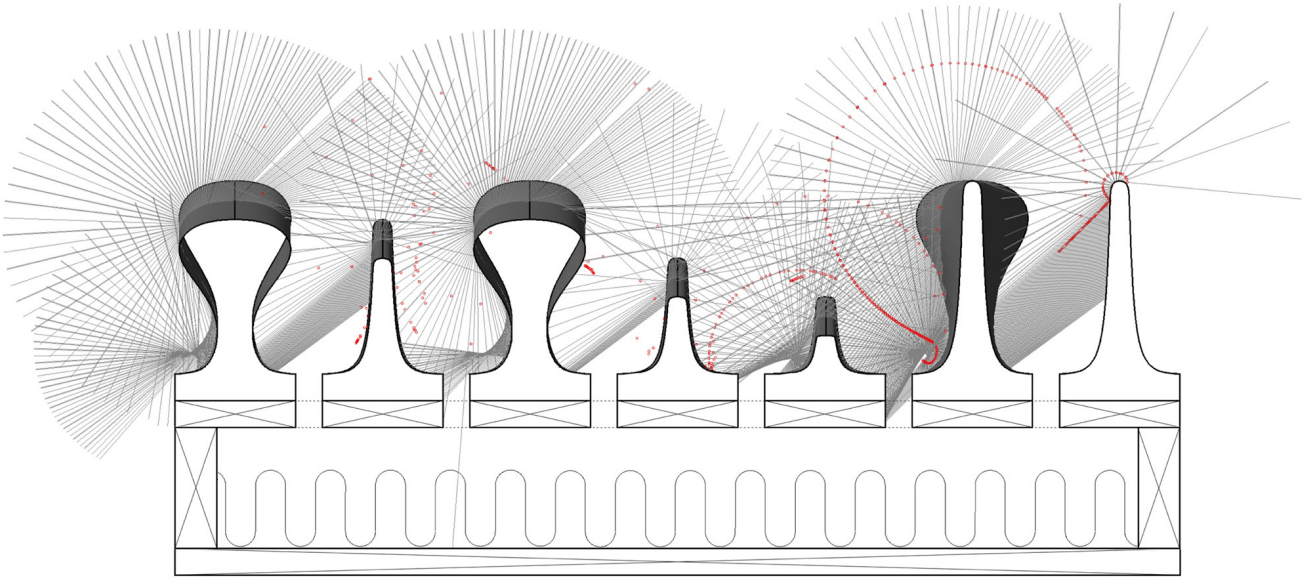
In this section we examine the added value of new synergies between digital design and manufacturing, which can not only achieve complex forms of architectural elements but also embed specific qualities through designed performance (Hensel and Menges 2008; Kolarevic 2005). We focus specifically on performance-driven ceramic bricks within the body of computational design research.

The most common performance criterion for bricks is naturally mechanical. This is exemplified by the early works at ETH Zurich of Gramazio and Kohler's programmed walls and columns, where robotic pick-and-place workflows were established, leveraging digital design information pipelines with the robot's capacity to precisely position bricks to produce bespoke statically stable curved walls (Bonwetsch 2012; Gramazio and Kohler 2008). Here, bricks are in their standard prismatic or perforated shape and are grabbed by either vacuum or spike end effectors and stabilized using glue applied by either a human operator or a static dispenser. In the subsequent iteration of the work, the focus has been on the development of a digital twin via lidar scanning to ensure the precision of the assembly (Dörfler 2018). A by-product of the systems' structural performance is an aesthetic quality, which is present in the robotically assembled facade of recycled bricks at the Chi She gallery (Archi-Union 2016) or in the Gantenbein Winery (Bonwetsch 2015). The latter not only aims to provide a playful "bubble" pattern to the prefabricated brick panels but also programs the space with specific requirements for shading and ventilation of the wine barrels.

While the above examples focus on the assembly of standard bricks, the following examples go beyond prismatic geometry towards free-form performance-driven shaping using various fabrication methods. The first is robotic static wire cutting. In the case of the Revolving brick project



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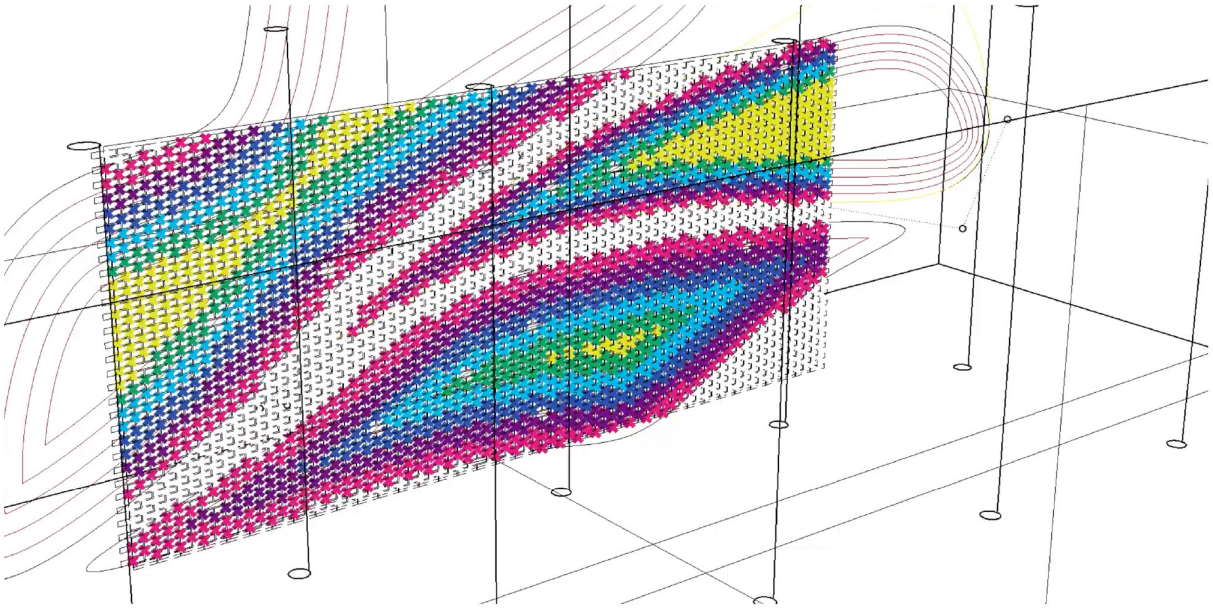


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2 The 14 families of acoustic-driven brick design (left to right), Fold and Cusp transitioning into flat tile, Cusp family, Cusp to Fold, and Fold family.

3 Cross-section showing the variable scattering qualities of the tiles and how their sequence creates complimentary interactions.

(Andreani and Bechtold 2014), wet clay bricks are cut using a robot-held end effector with a tensioned metal wire. This produces truncated volumes, which create a shading effect and presents a dynamic texturing of the façade. The geometries produced by this method are limited to shallow curvatures. 3D printing, on the other hand, allows for more geometric freedom in the brick, which is crucial for modulating thermal performance through porosity and mass distribution. This is exemplified through the work of Foged and Jensen (2018), where an energy model, thermal model, and fabrication model drive the shaping of 3D-printed thermal-based building blocks, where the mass distribution provides passive thermal effects for occupants. Another example is the Cool Brick (Rael and San Fratello 2015), which is a porous brick that absorbs water and slowly releases it into the interior environment as air moves through, providing a cooling effect via evaporation. Furthermore, the geometry of the brick is self-shading,



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4 Acoustic testing of the full scale demonstrator in Rhino using Pachyderm, with the implementation of a sound source and receiver setup.

which adds to the wall's performance. In contrast, the example of the PolyBrick H20 (Zhang, Qian, and Sabin 2019) uses the shape of the brick to retain water. In fact, the programmable microtexturing from the 3D printing, combined with a specific glazing technique, can steer the water flow on the surface of the brick.

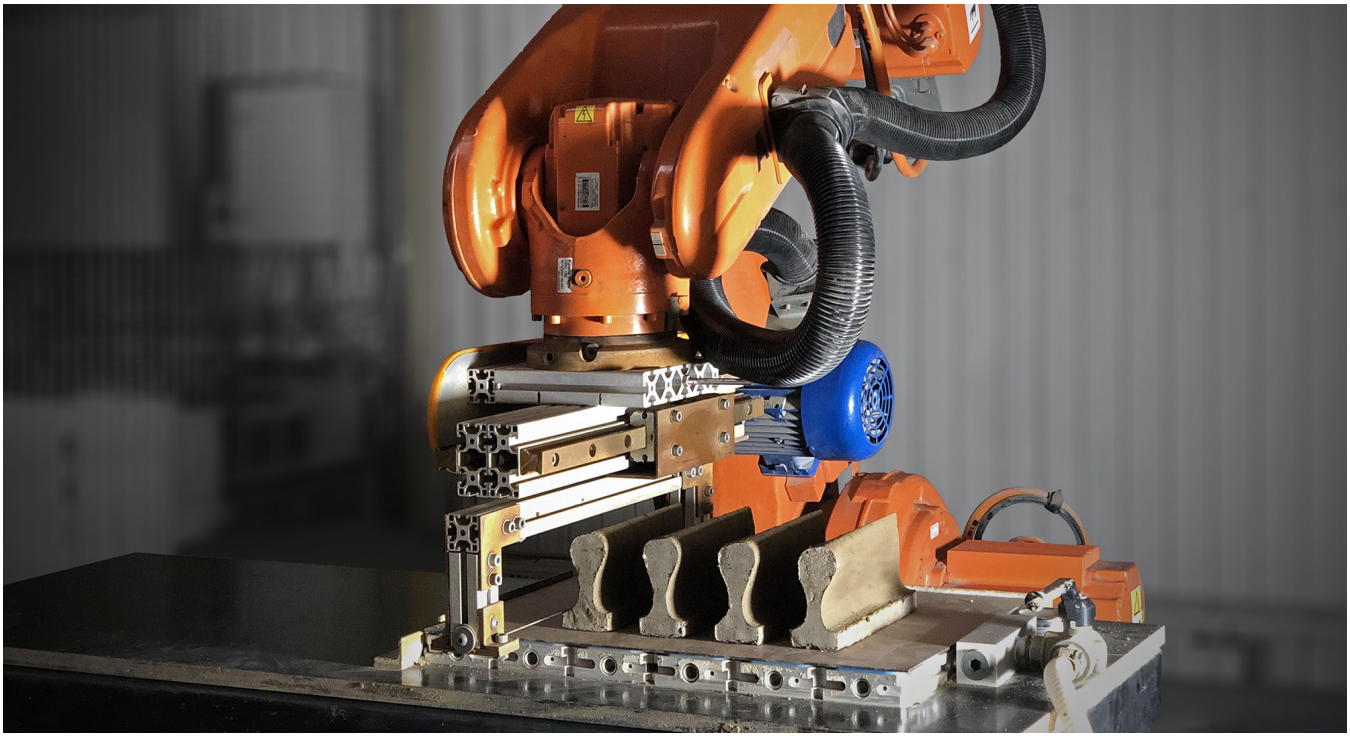
PROJECT BACKGROUND

Looking beyond the ceramic brick, digital simulation and fabrication have been used to create custom modules, arrayed in periodic patterns, to enhance acoustic performance of existing spaces. There are examples of full-scale demonstrators such as robotically pick-and-placed plastic extrusions (Vomhof et al. 2014) and robotically 3D-printed polyurethane (Bonswetch et al. 2008) or examples of 1:10 scaled samples such as robotically carved foam panels (Reindhart and Cabrera 2017) and 3D-printed hexagonal scattering patterns (Peters and Olesen 2010). While these projects primarily focus on the overall placement of materials, they do not address the specific design of the modular unit geometry. We believe that by combining the steering of local module geometry with the global pattern distribution, we can not only achieve a finer-tuned acoustic performance but also address pragmatic issues of fabrication and assembly.

In our intervention for the design of an acoustic partition to improve speech intelligibility in a university lecture hall, we design novel brick geometries driven by acoustic performance, specifically sound scattering, and limited by their manufacturing and assembly possibilities, specifically a ruled surface geometry and a grippable base. While the project could be considered a small-to-medium-scale intervention, it still comprised of 2,200 bricks arranged on a 30 m^2 wall. This marked the need for a comprehensive design-to-manufacturing workflow, beyond the lab setting. In fact, the only way to guarantee the humidity content of the bricks while cutting is by operating within the brick factory itself. We deploy our containerized robotic cell at the facilities of our industry partner, customizing our bricks directly on the production chain, as we detail below.

ACOUSTIC-DRIVEN BRICK DESIGN

The design research methods used are specifically focused on acoustics and include geometric, assembly, and spatial considerations. Initiating-- studies are conducted through geometric versioning paired with acoustic sound scattering properties, based on two primary forms, defined as Cusp and Fold surface definitions (McRobie 2017). The two base forms are selected from their different surface characteristics, with the Cusp defined by two opposing concave curves creating a pointy element, and the Fold defined by two opposing concave curves creating a convex broad curvature element (Fig. 2). Both geometries are maintained within the Danish Standard Brick dimensions (228 × 108 × 54 mm), which enable the elements to be nested as a modular brick system. Six variations of both Cusp and Fold geometries are then parametrically created to increase/decrease the geometric definition, also including two hybrids that combine the Cusp and Fold geometries.

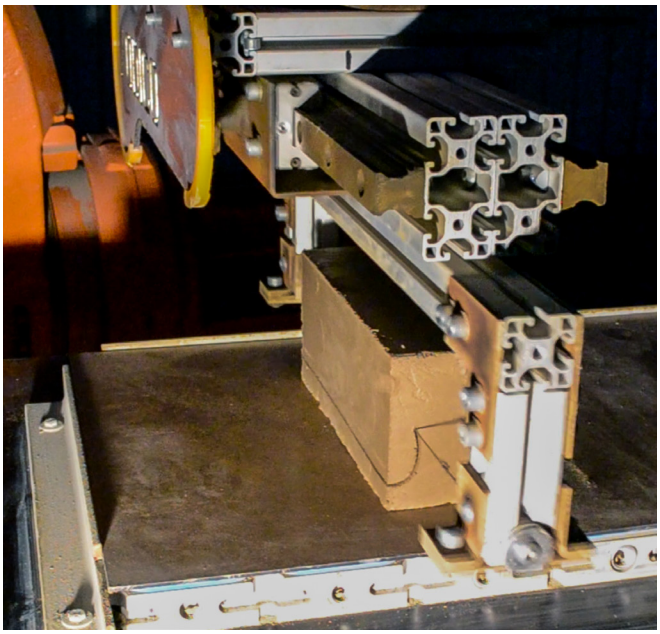


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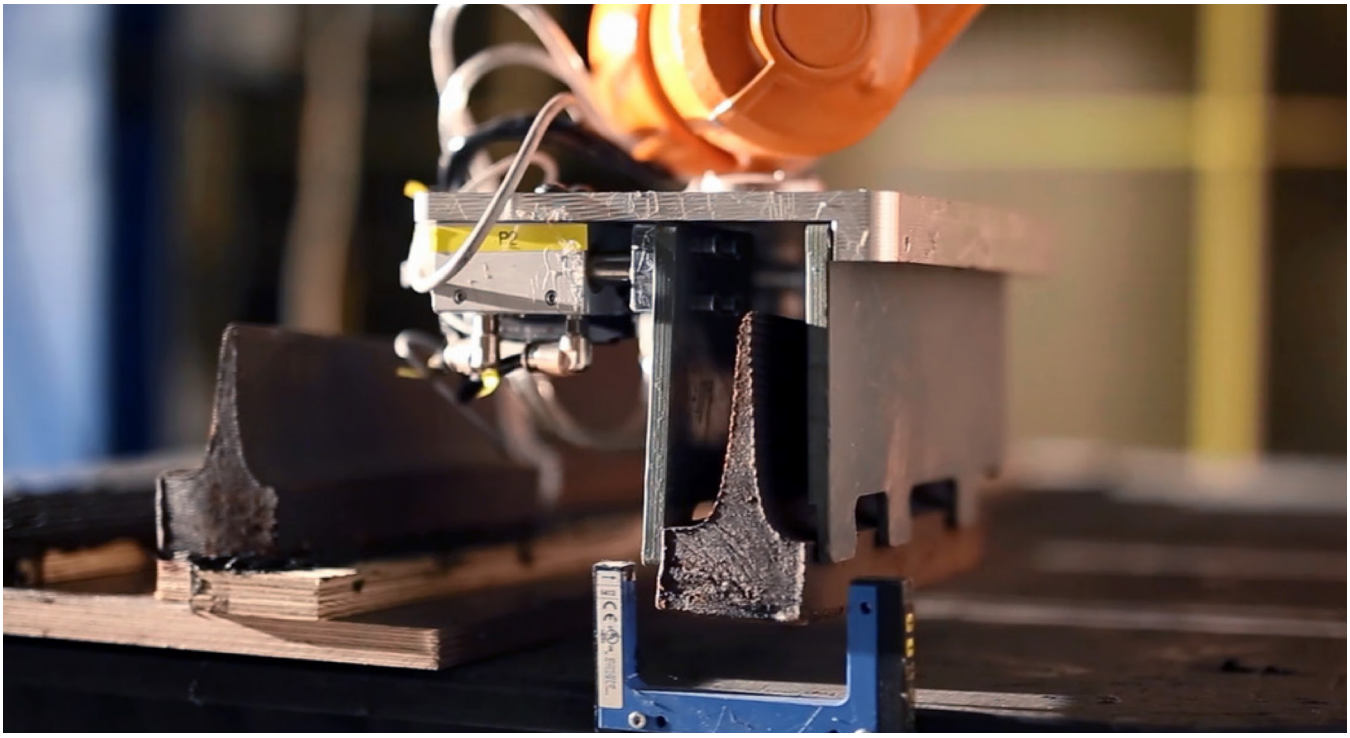
5 Robotic oscillating wire cutting (ROWC) station with the ABB 6620 deployed for production at our partner Strøjer Tegl.

6 Ribbed texture acquired by the cut brick, a unique characteristic of the ROWC (left). Cut bricks brought into the drying chambers (right).

7 ROWC's oscillating frame in the process of cutting a Fold family brick.

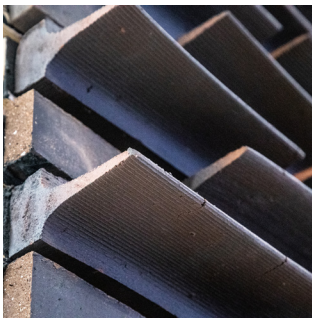
This creates a set of 14 unique yet combinable geometries, which enable making and studies of complex surfaces compositions that can be developed to act in respect to periodicity and modulation for specific sound frequencies (Cox and D'Antonio 2017).

Through a parametric design model, sound scattering of element compositions are then modeled with a specular ray-tracing method to understand the behavior of rays in relation to sectional configurations (Fig. 3). The alternation between Cusp and Fold allows the composition to steer the spreading of sound waves in relation to incidence wave-lengths, and thereby add design instrumentality across frequencies. With the tiles mounted on a base plate with perforations (that can be varied in the design model), sound energy can also be reflected into deeper structural layers containing porous material (stone wool insulation), allowing



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8 Robotic adaptive pick-and-place ABB 1600 station. An optical sensor adapts the tool path based on brick tolerance through online feedback.

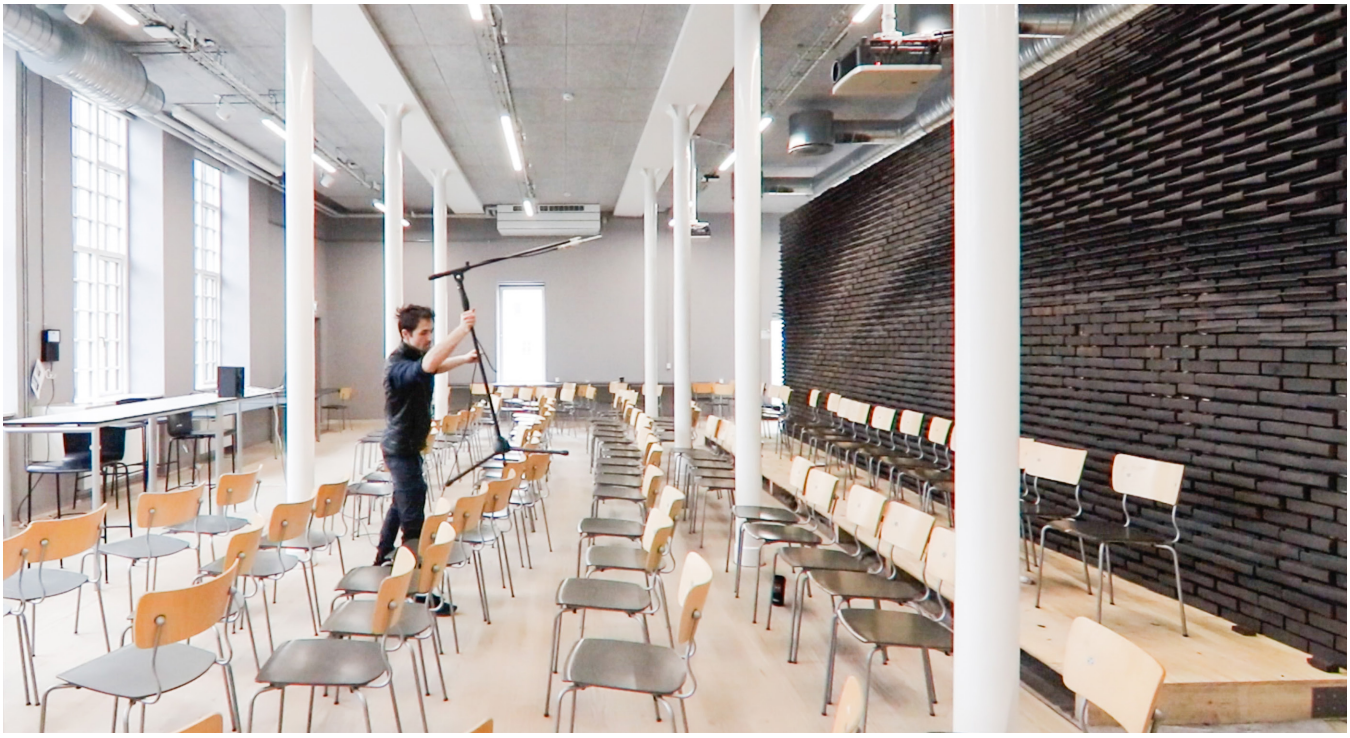
9 Close-up details of the assembled wall. The bricks are fixed with mortar onto CNC-milled plywood panels incorporating an acoustically absorbent cavity.

a dual acoustic design strategy of both scattering and absorption properties.

Similarly, sound scattering of different compositions are assessed by a finite volume method using Pachyderm Acoustical Simulation (van der Harten 2013), where simulated tile sample compositions span 2 × 2 m. Based on computational studies, combining topological versioning of Cusp and Fold geometries with sound scattering assessment through simulations, a final design model is developed, where the understanding of the sound spreading is embedded into the model where the designer intuitively can "sketch" design curves on a 30 m^2 canvas, to have the tiles distributed automatically according to acoustic and visual expressive characteristics (Fig. 4). In the built demonstrator, the pattern comprises a central vein of flat bricks and a gradient concentric distribution of the 14 families from flat to full height, alternating between Cusp and Fold families. This responds to a tile organization that aims to scatter sound at both higher and lower frequencies, and to illustrate the capacity of the tiles to create varied yet holistic articulations across the design system.

ROBOTIC BRICK MANUFACTURING

The acoustic-driven design of our bricks presents two challenges: the brick geometries are nonextrudable and the variation in their profile causes a different shrinkage ratio during firing, which means that the bricks come with a significant tolerance and are difficult to grip. We address these challenges by implementing novel robotic manufacturing workflows that build upon the state-of-the-art: robotic oscillating wire cutting and adaptive pick-and-place, which we describe below and in further detail in another publication (Rossi, Walker, and Søndergaard 2021).



10 The acoustic testing of the full-scale demonstrator that was measured with and without its presence to discern the effect.

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Initial Prototypes

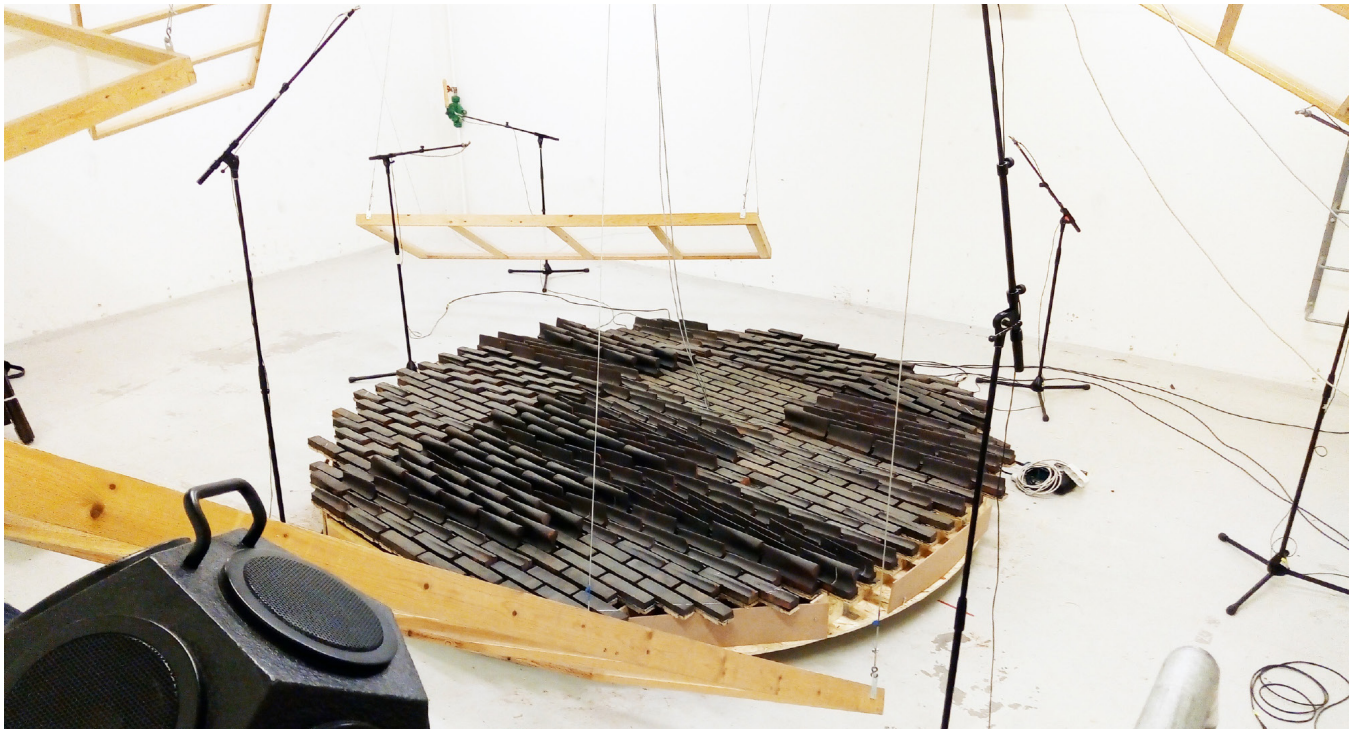
Early prototyping tests of Cusp-type bricks revealed problems with fabricating the complex geometries when considering the manufacturing chain of robotic-based oscillation cutting, drying, kiln burning, and transporting the sharp edges. Through iterative testing, the geometries were gradually adapted between both material and manufacturing performance with designed acoustic and visual properties. This required the inclusion of a minimum radius at the edges, without losing the intended geometry of the tiles.

Robotic Oscillating Wire Cutting (ROWC)

While static wire cutting of wet clay has proven effective for the truncated cutting of wet clay blocks, it has an exceptionally large failure rate when dealing with longer surface cuts, especially if they present tight curvatures and changes in direction like in our 14-brick family design. We have found that by introducing an oscillating motion to the wire, we are able to produce these geometries at a low failure rate and a high cutting speed of 4.5 mm/s (compared to 1 mm/s for a static wire). In fact, the reciprocating motion of the wire reduces the friction it encounters while cutting through the block of clay. By tuning the oscillation frequency, we can obtain a texture on the brick that also contributes to its acoustic properties (Fig. 6). We term the manufacturing process robotic oscillating wire cutting, and we design a proprietary end effector combining an electrical engine and wire frame. We deploy a robotic cell with an ABB 6620 (Figs. 5, 7) at our industrial partner facilities, where the bricks are cut in batches of 50 taken from the production chain. Fabrication time of the 2,000+ bricks, with the robot cell operated by two people, lasted 10 days. The cut bricks are left to dry in a humidity buffer zone for two weeks, then enter the drying chambers for a day and are subsequently taken on automatic carts into a tunnel kiln, where they are fired at 1200°C overnight.

Robotic Adaptive Pick-and-Place (APnP)

While our bricks have varying cross-sections, they share a common 25 mm high base, which is used by the gripper in assembly. We designed a custom finger-actuated gripper and attached it to an ABB 1600. The station also features a passive conveyor system for the different families that is manually restocked during the assembly process. An integrated computational workflow, from brick pattern design to robotic production, was developed in Rhino/Grasshopper to assemble 135 unique panels (Robert McNeel & Associates 2020; Rutten 2020). The outputs were ABB RAPID code informing the robot of the conveyance roller sequence from which it is to pick each brick and its ideal target location on the panel, which is adapted to brick tolerance using sensor feedback. As mentioned, the bricks come with a problematic varying tolerance of 2–8 mm due to differential shrinkage. Since it would be cumbersome for all 2,200 bricks to be scanned a priori and best fitted



11 Circular sample of the wall tested in a university lab.

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according to their tolerance, we developed a hands-off online feedback approach that embraces the tolerance of the bricks and adapts the previously generated robotic pick-and-place tool path to the real dimension of the brick read via an optical fork sensor (Fig. 8), on the fly without designer or robot operator intervention. The robot operator applies a mortar mix to the substrate panels, after which the robot commences the assembly, each panel taking 10 minutes.

On-Site Assembly

The 135 panels are designed with a carrying weight that is below 23 kg so they can be easily handled. The wall was assembled in the auditorium within half a day by 10 people. The substructure is composed of a platform reinforced with metal brackets, four structural timber columns, and stone wool insulation placed in between. The wall provides the room with a unique tectonic expression (Fig. 9).

VALIDATION OF ACOUSTIC PERFORMANCE

The acoustic tiles were subject to measurement both on site and in the laboratory. Spatial acoustic analysis (room analysis) was conducted in the lecture hall in which the demonstrator stood, following the ISO 3382:2008 acoustic room analysis standard. For a comprehensive acoustic analysis, a frequency sweep was conducted using the FuzzMeasure acoustic measurement software (RØDE Microphones 2020), with a focus on evaluating the reverberation time (T30) and definition (D50) of the lecture hall (Fig. 10). The general finding is that there is an improved speech intelligibility due to an increase of the definition. In the university laboratory reverberation test chamber, absorption and scattering measurements are conducted with a static 12 m^2 rectangular sample and a rotating circular sample, respectively, following the ISO methods for measurements (ISO:354:2003/2005) (Fig. 11). The measured samples are composed by a variation of tiles in segments of Cusp and Fold geometries to convey an understanding of the acoustic phenomena that can be created through such applications.

The results of the acoustic measurements in laboratory and field conditions illustrate that the tiles can be composed to scatter sound strategically in relation to the composition of the assembly, and in respect to specific frequencies. The results reflect the specific samples measured. It should be noted that the results are obtained in both laboratory and field conditions from planar compositions, which isolate sound scattering to be based entirely on the tile geometries and their collective planar distribution, resulting in higher scattering coefficients at higher frequencies. The scale and modularity of the tile system, however, enables the tiles to be mounted on a curved wall geometry, which theoretically will increase the sound scattering at lower frequencies due to longer wavelengths. The acoustic design and development could thereby enable an even greater variation in

sound scattering effects with such geometric additions to the design system.

CONCLUSION

This paper presented a novel integrated design-to-manufacture workflow for custom sound-scattering brick walls. We developed a simulation-based form-finding technique for performance-driven brick design, custom data management pipelines from design to production, and novel robotic fabrication methods, namely oscillating wire cutting and adaptive pick-and-place. We show the possibilities of using this workflow for improving sound qualities in buildings through the fabrication of a full-scale demonstrator, and we detail the acoustic results of our laboratory and field tests.

This project reveals the synergetic relations between material, form, fabrication technique, acoustics, and visual effects. These relations illustrate the necessity to investigate factors through an integrated design approach and to uncover interrelated features that contribute to a holistic and buildable design performance, as demonstrated. The strategic combination of modularity and in-element versatility is pivotal in allowing the acoustic variation properties, and thereby enabling the design system to be adapted to specific contexts, performances, and spatial intentions. While in this initial demonstrator we have focused our design effort on the brick family modules and their distribution on a planar substrate, future efforts will address the overall design of the partition. In fact, this system could be further extended by arraying the bricks over a curved substrate, which would contribute to an improved response in diffusing larger wavelengths at lower frequencies.

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IMAGE CREDITS

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